

Application Build and Release Automation Lab Assignment 1

Student Name: Viyom Rana
Roll Number: 2415001904

Scenario

A software company named GLA Digital Solutions is developing an Online Library Management System. The development team uses Linux and Git to manage project files and collaborate on application features.

Task 1 — Project Workspace

6 Marks

Create:

- **<FirstName>-GLA-LibrarySystem**

Inside it create:

- src
- docs
- config
- backup

Create: **library_info.txt**, including:

- Name
- Roll Number
- Subject
- Project Name
- Semester

Demonstrate Linux commands for:

- pwd
- directory navigation
- listing files
- displaying file contents
- displaying directory structure

Answer

GNU nano 8.7.1

library_info.txt

Name: Viyom Rana

Roll Number: 2415001904

Subject: Automation Lab

Project Name: F1-Forecast

Semester: 5th

[Wrote 5 lines]

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^ Replace

^K Cut

^U Paste

^T Execute

^J Justify

```
viyom@Viyom:~$ pwd
/home/viyom
viyom@Viyom:~$ cd
viyom@Viyom:~$ mkdir Viyom-GLA-LibrarySystem
viyom@Viyom:~$ cd Viyom-GLA-LibrarySystem
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ mkdir src docs config backup
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ touch library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ nano library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ cat library_info.txt
Name: Viyom Rana
Roll Number: 2415001904
Subject: Automation Lab
Project Name: F1-Forecast
Semester: 5th
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ tree
.
├── backup
├── config
├── docs
├── library_info.txt
└── src

5 directories, 1 file
viyom@Viyom:~/Viyom-GLA-LibrarySystem$
```

Task 2 — Git Repository

6 Marks

- Initialize Git.
- Configure Git username/email.
- Check status.
- Stage project files.
- Create the initial commit.
- Modify library_info.txt.
- Add a project objective and description.
- Check the changes.
- Stage the modified file.
- Create a second commit.
- Display commit history.

Answer

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git init
hint: Using 'master' as the name for the initial branch. This default br
hint: will change to "main" in Git 3.0. To configure the initial branch
hint: to use in all of your new repositories, which will suppress this w
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this com
mand:
hint:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/viyom/Viyom-GLA-LibrarySystem/
.git/
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git config --global user.name "Viy
om Rana"
error: invalid key: user.nameViyom Rana
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git config --global user.email "v
iyomanuragrana@gmail.com"
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to tr
ack)
```

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git config --global user.name
Viyom Rana
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git config --global user.email
viyomanuragrana@gmail.com
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git add library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git commit -m "Add library system
information"
[master (root-commit) be11149] Add library system information
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ |
```

Name: Viyom Rana

Roll Number: 2415001904

Subject: Automation Lab

Project Name: F1-Forecast

Semester: 5th

Project Objective: Forecast Winners

Description: It is an ML model

[Wrote 7 lines]

^G Help

^O Write Out

^F Where Is

^K Cut

^T Execute

^X Exit

^R Read File

^ Replace

^U Paste

^J Justify

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git add library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git commit -m "Add library system
information"
[master (root-commit) be11149] Add library system information
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git log --oneline
be11149 (HEAD -> master) Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ nano library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ cat library
cat: library: No such file or directory
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ cat library_info.txt
Name: Viyom Rana
Roll Number: 2415001904
Subject: Automation Lab
Project Name: F1-Forecast
Semester: 5th
Project Objective: Forecast Winners
Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git diff
diff --git a/library_info.txt b/library_info.txt
index 18613fc..d0fb629 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,3 +3,5 @@ Roll Number: 2415001904
  Subject: Automation Lab
  Project Name: F1-Forecast
  Semester: 5th
+Project Objective: Forecast Winners
+Description: It is an ML model
```

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git add library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git commit -m "Update library information"
[master 5dc4f08] Update library information
 1 file changed, 2 insertions(+)
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git log --oneline
5dc4f08 (HEAD -> master) Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem$
```

Task 3 — Feature Branch

6 Marks

Develop a Book Management Module.

Create branch: **book-feature**

Inside src, create: **book.txt**, describing the Book Management Module and containing at least three features.

- Stage and commit the changes.

Demonstrate:

- branch creation
- branch switching
- current branch
- all branches
- returning to main

Answer

Book Management Module

Purpose:

The Book Management Module is used to manage books in the online library>

Features:

1. Add new books to the library.
2. Search and view available books.
3. Update book information.
4. Remove books from the library.

Expected Users:

Students, faculty, and library staff.

[Read 13 lines]

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git branch book-feature
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git switch book-feature
Switched to branch 'book-feature'
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git branch
* book-feature
  master
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ git status
On branch book-feature
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem$ cd src
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ touch book.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano book.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano book.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat book.txt
Book Management Module
```

Purpose:

The Book Management Module is used to manage books in the online library system.

Features:

1. Add new books to the library.
2. Search and view available books.
3. Update book information.
4. Remove books from the library.

Expected Users:

Students, faculty, and library staff.

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add book.txt
```

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
```

On branch book-feature

Changes to be committed:

(use "git restore --staged <file>..." to unstage)
 new file: book.txt

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit -m "Add book management module"
```

[book-feature 79b154f] Add book management module

1 file changed, 13 insertions(+)

create mode 100644 src/book.txt

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --decorate
```

Task 4 — Merge Conflict

8 Marks

Merge book-feature into main.

Create: **library-update**

- Modify the same line in library_info.txt in library-update.
- Commit the change.
- Modify that same line differently in main.
- Commit the change.
- Attempt the merge.
- Identify the conflict.
- Resolve it manually.
- Stage and commit the resolved file.
- Display the final file.
- Display Git status and history.

Answer

```

viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --dec
orate
79b154f (HEAD -> book-feature) Add book management module
5dc4f08 (master) Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --dec
orate
79b154f (book-feature) Add book management module
5dc4f08 (HEAD -> master) Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git merge book-feature
Updating 5dc4f08..79b154f
Fast-forward
 src/book.txt | 13 ++++++++
 1 file changed, 13 insertions(+)
 create mode 100644 src/book.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat book.txt
Book Management Module

```

Purpose:

The Book Management Module is used to manage books in the online library system.

Features:

1. Add new books to the library.
2. Search and view available books.
3. Update book information.
4. Remove books from the library.

Expected Users:

Students, faculty, and library staff.

```

viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |

```

```

79b154f (HEAD -> book-feature) Add book management module
5dc4f08 (master) Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --dec
orate
79b154f (book-feature) Add book management module
5dc4f08 (HEAD -> master) Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git merge book-feature
Updating 5dc4f08..79b154f
Fast-forward
 src/book.txt | 13 ++++++++
 1 file changed, 13 insertions(+)
 create mode 100644 src/book.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat book.txt
Book Management Module

```

Purpose:

The Book Management Module is used to manage books in the online library system.

Features:

1. Add new books to the library.
2. Search and view available books.
3. Update book information.
4. Remove books from the library.

Expected Users:

Students, faculty, and library staff.

Name: Viyom Rana

Roll Number: 2415001904

Subject: Automation Lab

Project Name: F1-Forecast

Semester: 5th

Makes book management simple and efficient.

Description: It is an ML model

[Wrote 7 lines]

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^\ **Replace**

^K Cut

^U Paste

^T Execute

^J Justify

```

viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch library-update
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git switch library-update
Switched to branch 'library-update'
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
* library-update
  master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch library-update
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat ../library_info.txt
Name: Viyom Rana
Roll Number: 2415001904
Subject: Automation Lab
Project Name: F1-Forecast
Semester: 5th
Project Objective: To create a smart online library system that makes bo
ok management simple and efficient.
Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index d0fb629..039b14a 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001904
  Subject: Automation Lab
  Project Name: F1-Forecast
  Semester: 5th
-Project Objective: Forecast Winners
+Project Objective: To create a smart online library system that makes b
ook management simple and efficient.
  Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch library-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ../library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$

```

```
Subject: Automation Lab
Project Name: F1-Forecast
Semester: 5th
Project Objective: To create a smart online library system that makes book management simple and efficient.
Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index d0fb629..039b14a 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001904
  Subject: Automation Lab
  Project Name: F1-Forecast
  Semester: 5th
-Project Objective: Forecast Winners
+Project Objective: To create a smart online library system that makes book management simple and efficient.
  Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch library-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   ../library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch library-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   ../library_info.txt

viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit -m "Update library objective on library-update branch"
> git commit -m "Update library objective on library-update branch"
>
> ^C
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit -m "Update library objective on library-update branch"
[library-update 8977285] Update library objective on library-update bran
```

GNU nano 8.7.1

../library_info.txt

Name: Viyom Rana

Roll Number: 2415001904

Subject: Automation Lab

Project Name: F1-Forecast

Semester: 5th

Project Objective: To provide students and faculty with an easy-to-use

Description: It is an ML model

[Wrote 7 lines]

^G Help

^O Write Out

^F Where Is

^K Cut

^T Execute

^X Exit

^R Read File

^ Replace

^U Paste

^J Justify

```

* library-update
  master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat ../lllllllllllllibrary_inf
o.txt
cat: ../lllllllllllllibrary_info.txt: No such file or directory
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index 039b14a..d4350c5 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001904
  Subject: Automation Lab
  Project Name: F1-Forecast
  Semester: 5th
-Project Objective: To create a smart online library system that makes b
ook management simple and efficient.
+Project Objective: To provide students and faculty with an easy-to-use
online platform for managing library resources.
  Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit -m "Update library
objective on master branch"
[library-update 7d90b07] Update library objective on master branch
 1 file changed, 1 insertion(+), 1 deletion(-)
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git merge library-update
Already up to date.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |

```

```

1 file changed, 1 insertion(+), 1 deletion(-)
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git merge library-update
Already up to date.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --decorate -5
7d90b07 (HEAD -> library-update) Update library objective on master branch
8977285 Update library objective on library-update branch
79b154f (master, book-feature) Add book management module
5dc4f08 Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
* library-update
  master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git reset --hard HEAD~1
HEAD is now at 8977285 Update library objective on library-update branch
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ gti branch
Command 'gti' not found, did you mean:
  command 'gtg' from snap getting-things-gnome (0.6)
  command 'gtg' from deb gtg (0.6-12)
  command 'gt5' from deb gt5 (1.5.0~20111220+bzr29-4build1)
  command 'bti' from deb bti (034-6build4)
  command 'gth' from deb genomethreader (1.7.3+dfsg-11)
  command 'gt' from deb genomertools (1.6.6+ds-2)
  command 'gri' from deb gri (2.12.27-1.2build2)
  command 'gsi' from deb gambc (4.9.3-1.4build1)
  command 'ti' from deb ticgit (1.0.2.18-1)
  command 'gtf' from deb xserver-xorg-core (2:21.1.22-1ubuntu1)
  command 'gmi' from deb lieer (1.6-6)
  command 'git' from deb git (1:2.53.0-1ubuntu1)
See 'snap info <snapname>' for additional versions.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
  library-update
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |

```

GNU nano 8.7.1

../library_info.txt

Name: Viyom Rana

Roll Number: 2415001904

Subject: Automation Lab

Project Name: F1-Forecast

Semester: 5th

Go-use online platform for managing library resources.

Description: It is an ML model

[Wrote 7 lines]

^G Help

^O Write Out

^F Where Is

^K Cut

^T Execute

^X Exit

^R Read File

^ Replace

^U Paste

^J Justify

```

command 'gtg' from snap getting-things-gnome (0.6)
command 'gtg' from deb gtg (0.6-12)
command 'gt5' from deb gt5 (1.5.0~20111220+bzr29-4build1)
command 'bti' from deb bti (034-6build4)
command 'gth' from deb genomethreader (1.7.3+dfsg-11)
command 'gt' from deb genomertools (1.6.6+ds-2)
command 'gri' from deb gri (2.12.27-1.2build2)
command 'gsi' from deb gambc (4.9.3-1.4build1)
command 'ti' from deb ticgit (1.0.2.18-1)
command 'gtf' from deb xserver-xorg-core (2:21.1.22-1ubuntu1)
command 'gmi' from deb lieer (1.6-6)
command 'git' from deb git (1:2.53.0-1ubuntu1)
See 'snap info <snapname>' for additional versions.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
  library-update
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index d0fb629..d4350c5 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001904
  Subject: Automation Lab
  Project Name: F1-Forecast
  Semester: 5th
-Project Objective: Forecast Winners
+Project Objective: To provide students and faculty with an easy-to-use
online platform for managing library resources.
  Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commiy -m "Update library
  objective on master branch"
git: 'commiy' is not a git command. See 'git --help'.

The most similar command is
  commit
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$

```

```

command 'gt5' from deb gt5 (1.5.0~20111220+bzr29-4build1)
command 'bti' from deb bti (034-6build4)
command 'gth' from deb genomethreader (1.7.3+dfsg-11)
command 'gt' from deb genomertools (1.6.6+ds-2)
command 'gri' from deb gri (2.12.27-1.2build2)
command 'gsi' from deb gambc (4.9.3-1.4build1)
command 'ti' from deb ticgit (1.0.2.18-1)
command 'gtf' from deb xserver-xorg-core (2:21.1.22-1ubuntu1)
command 'gmi' from deb lieer (1.6-6)
command 'git' from deb git (1:2.53.0-1ubuntu1)
See 'snap info <snapname>' for additional versions.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
book-feature
library-update
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index d0fb629..d4350c5 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,5 +3,5 @@ Roll Number: 2415001904
Subject: Automation Lab
Project Name: F1-Forecast
Semester: 5th
-Project Objective: Forecast Winners
+Project Objective: To provide students and faculty with an easy-to-use
online platform for managing library resources.
Description: It is an ML model
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commiy -m "Update library
objective on master branch"
git: 'commiy' is not a git command. See 'git --help'.

The most similar command is
commit
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit
Aborting commit due to empty commit message.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |

```

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$
nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat ../library_info.txt
Name: Viyom Rana
Roll Number: 2415001904
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th

Project Objective: To provide students and faculty with an easy-to-use o
nline platform for managing library resources.
Project Description: The system helps users access, manage, and organize
library resources efficiently.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git diff
diff --git a/library_info.txt b/library_info.txt
index d0fb629..d87ffef 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -1,7 +1,8 @@
  Name: Viyom Rana
  Roll Number: 2415001904
-Subject: Automation Lab
-Project Name: F1-Forecast
+Subject: Application Build and Release Automation Lab
+Project Name: Online Library Management System
  Semester: 5th
-Project Objective: Forecast Winners
-Description: It is an ML model
+
+Project Objective: To provide students and faculty with an easy-to-use
online platform for managing library resources.
+Project Description: The system helps users access, manage, and organiz
e library resources efficiently.
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   ../library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |
```

GNU nano 8.7.1

../library_info.txt

Name: Viyom Rana

Roll Number: 2415001904

Subject: Application Build and Release Automation Lab

Project Name: Online Library Management System

Semester: 5th

<<<<<< HEAD

Project Objective: To provide students and faculty with an easy-to-use >

Project Description: The system helps users access, manage, and organiz>

=====

Project Objective: To create a smart online library system that makes b>

Description: It is an ML model

>>>>>> library-update

[Read 13 lines]

^G Help

^O Write Out

^F Where Is

^K Cut

^T Execute

^X Exit

^R Read File

^ Replace

^U Paste

^J Justify

```
no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ nano ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git add ../library_info.txt
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git commit -m "Resolve library information merge conflict"
[master 9a41459] Resolve library information merge conflict
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --decorate
9a41459 (HEAD -> master) Resolve library information merge conflict
6ef5053 Update library objective on master branch
8977285 (library-update) Update library objective on library-update branch
79b154f (book-feature) Add book management module
5dc4f08 Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ cat ../library_info.txt
Name: Viyom Rana
Roll Number: 2415001904
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th
```

Project Objective: To provide students and faculty with an easy-to-use online platform for managing library resources.

```
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git log --oneline --all --decorate
9a41459 (HEAD -> master) Resolve library information merge conflict
6ef5053 Update library objective on master branch
8977285 (library-update) Update library objective on library-update branch
79b154f (book-feature) Add book management module
5dc4f08 Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$
```

Task 5 — Submission

4 Marks

Submit:

- Screenshots of important commands.
- Project directory.
- Git history.
- Branch information.
- Merge demonstration.
- Conflict and resolution demonstration.

Answer

```
ch
79b154f (book-feature) Add book management module
5dc4f08 Update library information
be11149 Add library system information
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git branch
  book-feature
  library-update
* master
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-LibrarySystem/src$ |
```